Homework (Rotation and Transformation matrices) (110 marks)

Proofs

1. In your own words, prove using trignometry that the 2D rotation matrix is given by (10 marks)

$$R = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

2. Derive the expression for Euler angles roll, pitch yaw from a given 3D rotation matrix (10 marks)

3. Derive the rodrigues formula for a rotation matrix that rotates a point around a given unit vector k for an angle theta. (20 marks)

4. Derive the axis-angle representation from a given 3D rotation matrix (10 marks)

5. Given any 3x3 matrix A, how would you test for it to be a valid rotation matrix? You have to test for two properties. What are those two properties? (10 marks)

6. Find the inverse of a 4x4 homegenous Transformation matrix

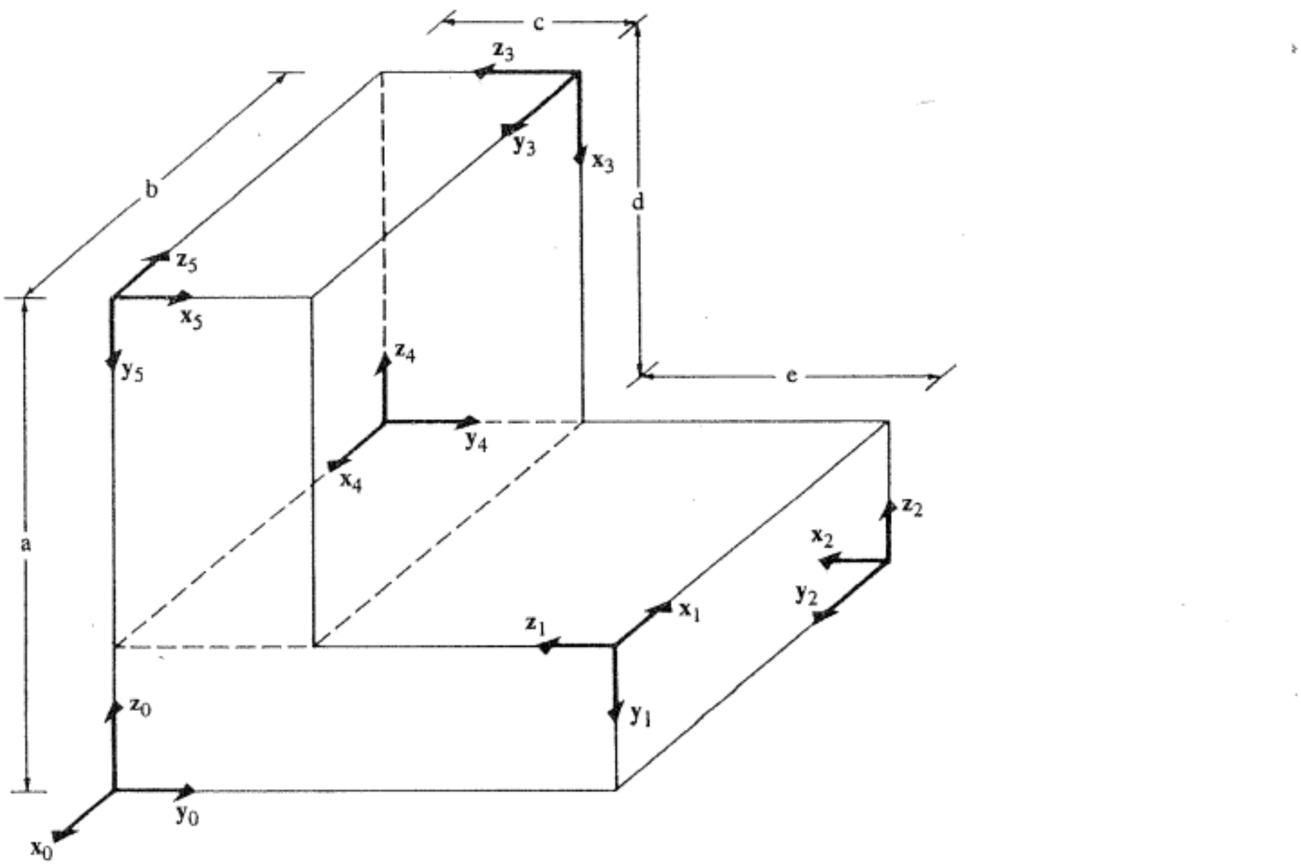
   T = [[R, t],
   [0, 1]]

   Assume R to be a valid Rotation matrix. (10 marks)

7. If the OUVW system has basis vectors $\mathbf{u} = (1/\sqrt{2}, 0, 1/\sqrt{2})^T$, $\mathbf{v} = (-1/\sqrt{2}, 0, 1/\sqrt{2})^T$, $\mathbf{w} = (0, -1, 0)^T$, and the OXYZ system has basis vectors $\mathbf{x} = (1, 0, 0)^T$, $\mathbf{y} = (0, 1/\sqrt{2}, -1/\sqrt{2})^T$, $\mathbf{z} = (0, 1/\sqrt{2}, 1/\sqrt{2})^T$, then what is the corresponding rotation matrix between the two systems? (10 marks)

8. You can use relationships between Basis vectors to find Rotations and translations between two coordinate frames. (16 marks)

2.6 For the figure shown below, find the 4×4 homogeneous transformation matrices ${}^{i-1}\mathbf{A}_i$ and ${}^0\mathbf{A}_i$ for $i = 1, 2, 3, 4, 5$.



For example, $z_1 = -y_0$, $x_1 = -x_0$, $y_1 = -z_0$, means that you can write $B_1 = [x_1, y_1, z_1] = [-x_0, -z_0, -y_0]$.

Assuming $B_0 = I$ to be identity, $x_0 = [1, 0, 0]$, $y_0 = [0, 1, 0]$, $z_0 = [0, 0, 1]$.

Hence, $B_1 = [-\mathbf{x}_0 \quad -\mathbf{z}_0 \quad -\mathbf{y}_0] = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

Now we can get rotation using, ${}^0R_1 = B_0^T B_1 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

Also, translation along y_0 = (c+e), hence translation ${}^0\mathbf{t}_1$ = [0, c+e, 0].
Therefore, 4x4 homogeneous matrix is

$$
{}^0A_1 = \begin{bmatrix} {}^0R_1 & {}^0\mathbf{t}_1 \\ \mathbf{0}^\top & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & c+e \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

9 Find the 7 transformation matrices (14 marks)

2.7 For the figure shown below, find the 4 × 4 homogeneous transformation matrices ${}^{i-1}\mathbf{A}_i$ and ${}^0\mathbf{A}_i$ for i = 1, 2, 3, 4.